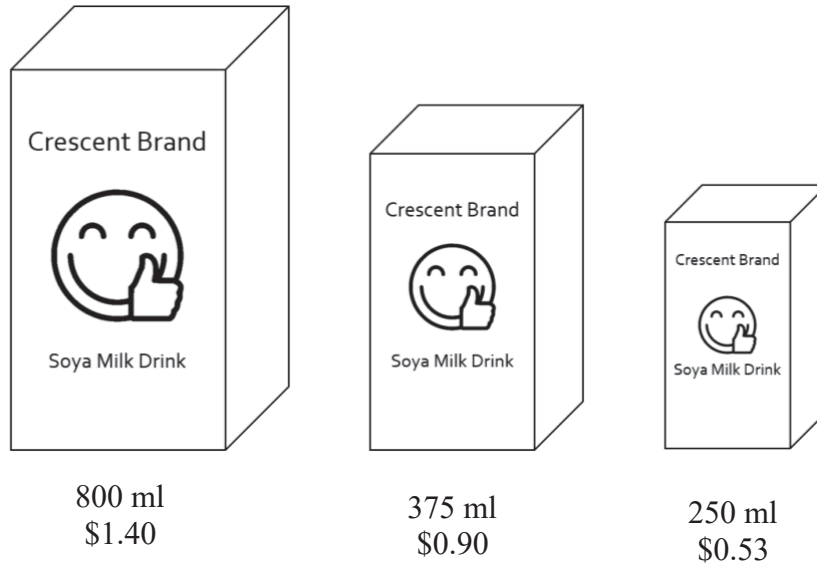


- 22 A company manufactures three sizes of the same brand of soya milk.



- (a) Show that the cost of the soya milk is not directly proportional to the quantity of soya milk.

Answer

[2]

- (b) The packets are geometrically similar.
The height of the 250 ml packet is 11 cm.

Calculate the height of the 800 ml packet.

Answer cm [3]

END OF PAPER

ANSWER KEY

Qn		Qn									
1	$k = 1$	15a	999.8 g								
2	71	15b	9.00 g								
3a	2 days	15c	For grouped data, only the mid-values of each data are used to calculate the mean and standard deviation. Hence, the mean and standard deviation are estimates.								
3b	0.25	15d	I will buy machine A. Machine A is more consistent in producing the packets of 1kg chocolate than machine B because the standard deviation of the mass of chocolate by machine A is 9.00g which is lower than the standard deviation of the mass of chocolate by machine B that is 12.5g.								
4	Quadratic Sketch	16a	$2 \times 3 \times 5^2$								
5a	2 m/s	16b	$a = 906$								
5b	0 m/s^2	17ai	$\begin{pmatrix} 117 \\ 96 \end{pmatrix}$								
5c	4 m/s	17aai	The elements represent the manufacturing cost of a dress (\$117) and a shirt (\$96) respectively.								
5d	Speed-Time graph	17bi	(26100)								
6a	Venn Diagram	17bii	The total manufacturing cost of 100 dresses and 150 shirts.								
6b	$\{11, 13, 17, 19\}$	18a	$(2a - b + 7)^2$								
6c	10	18b	$(y - 3)(xy + 3x - y)$								
7	212.5 ml	19a	<table border="1"> <tr> <td>4</td> <td>2, 3, 4, 5,</td> <td>23</td> <td>25</td> </tr> <tr> <td></td> <td>4, 3, 2</td> <td></td> <td></td> </tr> </table>	4	2, 3, 4, 5,	23	25		4, 3, 2		
4	2, 3, 4, 5,	23	25								
	4, 3, 2										
8a	$n = 6$	19b	$k = 9$								
8b	20°	19c	$S + 2 = (n + 1)^2$ or $S = (n + 1)^2 - 2$								
9	Sketch	19d	9799								
10	Construction	20	$m = -38$ $x = 8$								
11	$\frac{x-1}{x+1}$	21a	$c = \frac{6a+5b}{2(a+b)}$								
12	8 : 7	21b	$c = 4$								
13	$r = 3.00$	22a	Since the cost per ml for the 800 ml packet is different from that of the 375 ml packet and the 250 ml packet, the cost of the soya milk is not directly proportional to the quantity of soy milk.								
14a	48°	22b	16.2 cm								
14b	96°										
14c	48°										
14d	6°										